Piter Z. Garcia Bautista

Machine Learning, Quantum Computing, and Reproducible Quantum Systems Research

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Summary

Graduate researcher with an M.S. in Computer Science and current graduate work in Data Science at Rochester Institute of Technology. Brings industry experience building reliable data and machine learning systems together with current AI and quantum research focused on reproducible evaluation, quantum routing, resource allocation, and learning under stochastic and adversarial conditions. Research interests center on structured machine learning for reliable quantum systems, robust experimentation, and decision-making from noisy and limited observations.

Technical Skills

Programming: Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash
Machine Learning / AI: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, contextual bandits, neural bandits, supervised learning, model evaluation, error analysis
Data Science / Statistics: pandas, NumPy, SciPy, Matplotlib, Plotly, Seaborn, reproducible benchmarking, experiment design, applied statistics
Quantum / Research Focus: quantum routing, qubit allocation, stochastic and adversarial evaluation, multi-run validation, reliability-oriented benchmarking
Research Tooling: Git/GitHub, Linux, Docker, LaTeX, AWS, GCP

Relevant Experience

Graduate Assistant – Quantum & AI Research Fall 2025 – Present
Software Engineering Department, Rochester Institute of Technology *Rochester, NY*

- Build reproducible multi-testbed evaluation workflows for quantum routing and resource-allocation studies across local, Colab, and GCP environments, with shared datasets, structured logging, automated validation, and comparison-ready outputs.
- Develop Python-based experimental systems for contextual bandits, adversarial neural bandits, and dynamic qubit-allocation analysis under stochastic and non-stationary quantum-network conditions.
- Produce technical reports and manuscript-ready analyses that compare algorithms across attack scenarios, allocation strategies, and capacity regimes, emphasizing robustness, interpretability, and disciplined evaluation.

University of Rochester, Warner School of Education Sep 2024 – Present
CS Teacher Candidate & Education Research (NSF Noyce Scholar) *Rochester, NY*

- Deliver K–12 computer science instruction in active public school placement as part of the NSF Robert Noyce Scholar program, including computational thinking, inclusive pedagogy, and NYS learning standards.
- Conduct applied education research on AI-informed and data-driven approaches to improve learning outcomes for neurodivergent students, applying Universal Design for Learning (UDL) and evidence-based adaptive strategies.
- Produce lesson artifacts and instructional analyses documenting the intersection of AI, inclusive education, and evidence-based CS pedagogy.

AI Data Solutions Engineer Jan 2021 – Sep 2022
VIOME Inc. *New York, NY*

- Developed applied AI and machine learning workflows in Python for healthcare-oriented use cases, including preprocessing, feature engineering, model experimentation, and evaluation.
- Worked in AWS-backed environments to support reproducible data preparation and reliable evaluation for predictive and recommendation-oriented applications.

Data Solutions Engineer Sep 2022 – Aug 2024
VEDADATA Inc. *New York, NY*

- Designed large-scale Python and pandas analytics workflows for ingestion, cleaning, preprocessing, validation, and downstream analysis.
- Applied data-quality checks and statistical reasoning to improve traceability, reliability, and usability in production data systems.

Education

Master of Science in Data Science Expected Aug 2026
Rochester Institute of Technology *Rochester, NY*

- Relevant coursework: Data Science Foundations, Software Engineering for Data Science, Data-Driven Knowledge Discovery, Neural Networks, Applied Statistics, Bioinformatics Algorithms.

Master of Science in Computer Science 2015
Rochester Institute of Technology *Rochester, NY*
- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics.

Additional Graduate Study: M.S. in Teaching Computer Science K–12 Expected Aug 2026
University of Rochester, Warner School of Education *Rochester, NY*
- NSF Robert Noyce Scholar; Advanced Certificate in Leadership in Disability and Inclusive Practices.

Selected Technical Writing

- *Quantum Path Optimization with Fault-Tolerant Routing* — manuscript-level analysis of quantum-classical hybrid ML for fault-tolerant routing under noise constraints (under review, 2025–2026).
- *Big Data Medical Diagnosis: A Multi-Method Approach* (RIT M.S. Computer Science graduate research, 2015) — ensemble and deep learning methods for clinical classification on high-dimensional biomedical data.
- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report, 2025) — ensemble and regularized regression methods for clinical risk stratification under distribution shift.
- *BIO614-FinalProjectProposal* (Overleaf manuscript, 2026) — RNA secondary structure prediction with thermodynamic deep learning enhancements, reproducible benchmarking, and biological interpretation.
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025) — fairness-aware ML for genomic diagnostic algorithms with SHAP-based bias detection and statistical significance testing.
- *Differential Gene Expression in Murine DRG Neurons Following Sciatic Nerve Injury: An NGS Reanalysis* (BIOL550 Computational Biology Project, 2026) — reproducible bulk RNA-seq pipeline using DESeq2, QC validation, and pathway-level biological interpretation of nociceptor gene expression.
- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Proposal, 2025) — bandit learning with equity constraints applied to high-stakes clinical diagnostic decision-making, addressing demographic disparity in sequential clinical recommendation workflows.

Selected Fit for This Role

- Current graduate assistant research on quantum routing, dynamic qubit allocation, contextual bandits, and adversarial neural bandits with reproducible multi-run evaluation across noisy and changing conditions.
- Strong Python-based machine learning and experiment-design background aligned with learning from limited observations, robust benchmarking, and careful analysis of reliability tradeoffs in quantum settings.
- Clear motivation to extend current quantum-systems ML work toward structured learning for reliable quantum computing, including noise-aware modeling and error-mitigation research in the KTH/WASP environment.

References

- Professor Travis Desell, Rochester Institute of Technology – advisor and reference for data science, machine learning, and computational research readiness.
- Dr. Daniel Krutz, Rochester Institute of Technology – reference for technical rigor, software engineering, and research preparation.
- Additional reference details available on request.